

# Swasthya Health Insurance — Dummy Policy Dataset

## Generation Process, Attribute Definitions, and Coverage Analytics

**Dataset:** 20 insurance products · 67 plan tiers · 75 typed attributes · fictional insurer (Swasthya Health Insurance Co., IRDAI Reg. 147) **Purpose:** RAG knowledge base for an AI voice agent demo **Files generated:** (swasthya\_SP001.txt) - (swasthya\_SP020.txt) + (policy\_metadata.json) + (policy\_feature\_registry.py)

### 1. Generation Process

#### 1.1 Design Principles

The dataset was designed to be internally consistent across three dimensions:

**Premium ordering** — at equivalent age and SI, premiums follow a strict hierarchy: (SP004 < SP001 < SP002 < SP003) and (SP009 ≈ 45% below SP001) for young adults, (SP006 significantly above SP001–SP003) for seniors.

**Regulatory realism** — all waiting periods, exclusion lists, AYUSH provisions, and Arogya Sanjeevani terms mirror actual IRDAI circulars. UINs follow the (SWIHLIP24XXXV012024) format. Section 80D limits, PED definitions, and portability rights are accurate to Indian insurance regulation as of 2024.

**Coverage diversity** — products span 8 distinct primary need segments, 4 buyer types, 4 budget bands, and entry ages from 18 to 80. No two products are interchangeable.

#### 1.2 Product Architecture

Each of the 20 (.txt) policy documents follows an 11-clause structure:

| Clause | Content                                                         |
|--------|-----------------------------------------------------------------|
| 1      | Definitions — typed vocabulary for all terms used in the policy |
| 2      | Plan options and premium schedule                               |
| 3      | Scope of cover / worked examples                                |
| 4      | Detailed benefit provisions                                     |
| 5      | Waiting periods                                                 |
| 6      | Exclusions                                                      |
| 7      | Claims procedure                                                |
| 8      | Renewal and NCB                                                 |
| 9      | Eligibility and underwriting                                    |
| 10     | Tax / regulatory information                                    |
| 11     | Product-specific feature (unique to each product)               |

### 1.3 The Feature Registry

Every attribute from every policy document was extracted and typed into `policy_feature_registry.py` — a single Python dict of dicts that is the ground truth for all downstream logic. The registry uses five types:

- `bool` — feature present (`True`) or absent (`False`)
- `int` (INR) — monetary amounts in Indian Rupees
- `float` (pct) — percentages expressed as 0.0-1.0
- `str_enum` — value from a defined set (e.g. `"once_different_illness"`)
- `dict` — nested sub-attributes (waiting periods, NCB, plans, risk tiers)

`None` means not applicable — distinct from `False` (absent).

### 1.4 The Attribute Glossary

`attribute_glossary.py` defines 75 attributes across three layers:

| Layer  | Count | Purpose                                                |
|--------|-------|--------------------------------------------------------|
| User   | 16    | Fields collected from the caller during info-gathering |
| Policy | 50    | Product-level attributes in the registry               |
| Plan   | 9     | Plan-tier attributes (SI, premium, OPD limit, etc.)    |

Each entry includes: canonical key, type, label, LLM-ready description, valid values (for enums), unit, nullable flag, hard-filter flag, ask order, and question text.

## 2. The 20 Products

| Code  | Name                        | Segment                       | Plans | Premium Range     | Entry Age |
|-------|-----------------------------|-------------------------------|-------|-------------------|-----------|
| SP001 | Swasthya Protect            | Entry-level individual/family | 3     | ₹4,400 - ₹26,500  | 18-65     |
| SP002 | Swasthya Protect Plus       | Mid-tier comprehensive        | 4     | ₹7,800 - ₹82,000  | 18-65     |
| SP003 | Swasthya Protect Global     | Premium international         | 3     | ₹32,000 - ₹98,000 | 18-65     |
| SP004 | Swasthya Arogya Sanjeevani  | IRDAI standardised            | 3     | ₹1,800 - ₹5,100   | 18-65     |
| SP005 | Swasthya Micro Bima         | Micro / low-income            | 2     | ₹799 - ₹1,199     | 18-60     |
| SP006 | Swasthya Senior Shield      | Senior citizens               | 4     | ₹18,000 - ₹98,000 | 60-80     |
| SP007 | Swasthya Maternity Suraksha | Maternity (female)            | 2     | ₹13,800 - ₹20,200 | 21-45     |
| SP008 | Swasthya Women Wellness     | Women-specific                | 2     | ₹13,200 - ₹24,500 | 25-55     |
| SP009 | Swasthya Young Star         | Youth / student               | 2     | ₹3,400 - ₹5,800   | 18-35     |
| SP010 | Swasthya Super Top-Up       | Aggregate top-up supplement   | 6     | ₹3,400 - ₹8,200   | 18-65     |

| Code  | Name                              | Segment                            | Plans | Premium Range       | Entry Age        |
|-------|-----------------------------------|------------------------------------|-------|---------------------|------------------|
| SP011 | Swasthya Hospital Daily Cash      | Fixed daily cash benefit           | 4     | ₹1,500 - ₹7,200     | 18-65            |
| SP012 | Swasthya Personal Accident Shield | Accident protection                | 4     | ₹550 - ₹4,000       | 18-60            |
| SP013 | Swasthya Critical Guard           | Critical illness lump-sum          | 3     | ₹5,800 - ₹25,800    | 21-60            |
| SP014 | Swasthya Cancer Protect           | Cancer stage-based benefit         | 3     | ₹11,800 - ₹24,600   | 21-60            |
| SP015 | Swasthya Heart + Diabetes Care    | Chronic disease (diabetes/cardiac) | 3     | ₹18,500 - ₹44,000   | 25-65            |
| SP016 | Swasthya Corporate Care           | Group health (50+ employees)       | 4     | ₹12,000 - ₹52,000   | No age limit     |
| SP017 | Swasthya SME Suraksha             | Group health (7-50 employees)      | 3     | ₹1,200 - ₹2,800     | No age limit     |
| SP018 | Swasthya AI Health Companion      | AI/wearable integrated             | 3     | ₹11,800 - ₹46,000   | 21-55            |
| SP019 | Swasthya Gig Worker Shield        | Gig/informal economy               | 5     | ₹1,095 - ₹3,299     | 18-60            |
| SP020 | Swasthya Family Legacy            | Multi-generational (3 generations) | 4     | ₹62,000 - ₹4,20,000 | 25-55 (proposer) |

**Total: 20 products • 67 plan tiers**

### 3. Attribute Definitions, Coverage, and Filtering Priority

Attributes are grouped into six categories. **Filtering priority** indicates how early the attribute should be used to eliminate or rank products:

- **Hard filter** — eliminates products immediately when mismatched; ask first
- **Strong soft signal** — high scoring weight; ask before retrieval
- **Moderate signal** — improves ranking quality; ask after minimum fields
- **Plan-level** — selects plan tier within confirmed product; ask last
- **Universal / low differentiator** — present in most or all products; rarely decisive

### 3.1 Eligibility Attributes

These are hard filters — if a value is known and mismatched, the product is eliminated regardless of other scores.

| Attribute                  | Type                                | Definition                                                                                                                                                                                                                                                                                       | Products Covered | Priority                                                                                       |
|----------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------|
| <code>buyer_type</code>    | <code>str_enum</code>               | Who is purchasing: <code>individual</code> , <code>employer_large</code> (50+ employees), <code>employer_sme</code> (7-50 employees), <code>gig_worker</code> . The single most powerful filter — <code>employer_large</code> resolves to SP016 alone; <code>employer_sme</code> to SP017 alone. | All 20           |  Ask 1st    |
| <code>age</code>           | <code>int</code><br>(years)         | Applicant's age. Determines which products are available: SP006 requires 60–80; SP009 caps at 35; SP007/SP008 cap at 45/55; most others accept 18–65. Also drives premium pricing within a product.                                                                                              | All 20           |  Ask 2nd    |
| <code>gender</code>        | <code>str_enum</code>               | <code>female</code> , <code>male</code> , or <code>other</code> . Female unlocks SP007 (Maternity) and SP008 (Women Wellness); male immediately excludes those two. Also affects pricing on SP013/SP014 (female gets 10% discount).                                                              | All 20           |  Ask 3rd  |
| <code>entry_age_min</code> | <code>int</code><br>(years, policy) | Minimum age for first purchase. Ranges: 18 (most), 21 (SP007/SP013/SP014/SP018), 25 (SP008/SP015/SP020), 60 (SP006).                                                                                                                                                                             | All 20           |  Registry |
| <code>entry_age_max</code> | <code>int</code><br>(years, policy) | Maximum age for first purchase. Ranges: 35 (SP009), 45 (SP007), 55 (SP008/SP018), 60 (SP012/SP013/SP014), 65 (most), 80 (SP006). None for group products.                                                                                                                                        | 17/20            |  Registry |

Age gate summary:

| Age band | Products eligible for new entry                            |
|----------|------------------------------------------------------------|
| 18-20    | SP001-SP005, SP009-SP012, SP016-SP017, SP019 (15 products) |
| 21-24    | + SP007, SP013, SP014, SP018 (19 products)                 |
| 25-35    | All 20 products                                            |
| 36-59    | All except SP009 (19 products)                             |
| 60-65    | SP001-SP006, SP010, SP015-SP017, SP020 (12 products)       |
| 66-80    | SP006 only (renewal only for others)                       |

3.2 Need / Intent Attributes

These directly point to specific products. Seven of the eight `primary_need` values resolve to exactly one product.

| Attribute                 | Type                  | Definition                                                                                                                                                                                                                                                                                                                                                                                    | Products Covered | Priority              |
|---------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------|
| <code>primary_need</code> | <code>str_enum</code> | The main insurance need. Values and their product mapping:<br><code>hospitalisation</code> → 14 products;<br><code>maternity</code> → SP007 only; <code>cancer</code> → SP014 only; <code>critical_illness</code> → SP013 only; <code>accident</code> → SP012 only;<br><code>top_up</code> → SP010 only;<br><code>international</code> → SP003 only;<br><code>daily_cash</code> → SP011 only. | All 20           | 🔴 Ask 4th             |
| <code>has_ped</code>      | <code>bool</code>     | Whether the applicant has any pre-existing diagnosed condition. <code>True</code> triggers <code>ped_type</code> follow-up. Affects waiting period selection across all 18 indemnity products.                                                                                                                                                                                                | 18/20            | 🟡 Ask 5th             |
| <code>ped_type</code>     | <code>str_enum</code> | Type of pre-existing disease:<br><code>diabetes_cardiac</code> (strongest signal — points to SP015 which has 12-month wait vs 36-48 elsewhere), <code>other_ped</code> , or <code>none</code> . Only relevant if <code>has_ped=True</code> .                                                                                                                                                  | 18/20            | 🔴 Ask 5th (follow-up) |
| <code>needs_opd</code>    | <code>bool</code>     | Whether the user wants outpatient (non-hospitalisation) cover for doctor visits and medicines. Eliminates SP001, SP004, SP005, SP006, SP010, SP011, SP012, SP013, SP014 (9 products) when <code>True</code> .                                                                                                                                                                                 | 11/20 have OPD   | 🟡 Ask 6th             |

**Primary need → product mapping:**

| Primary need     | Products eligible                             | Products eliminated       |
|------------------|-----------------------------------------------|---------------------------|
| hospitalisation  | SP001-SP006, SP008-SP009, SP015-SP020<br>(14) | SP007, SP010-SP014<br>(6) |
| maternity        | SP007 (1)                                     | All others (19)           |
| cancer           | SP014 (1)                                     | All others (19)           |
| critical_illness | SP013 (1)                                     | All others (19)           |
| accident         | SP012 (1)                                     | All others (19)           |
| top_up           | SP010 (1)                                     | All others (19)           |
| international    | SP003 (1)                                     | All others (19)           |
| daily_cash       | SP011 (1)                                     | All others (19)           |

### 3.3 Affordability Attributes

| Attribute     | Type     | Definition                                                                                                                                                                                                                                                                   | Products Covered | Priority  |
|---------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------|
| budget_band   | str_enum | Annual premium tolerance: <b>micro</b> (<₹2,000 — SP005, SP019), <b>budget</b> (₹2K-₹10K — 7 products), <b>mid</b> (₹10K-₹30K — 12 products), <b>premium</b> (>₹30K — 6 products). Acts as guardrail, not hard filter — budget mismatch reduces score but doesn't eliminate. | All 20           | 🟡 Ask 7th |
| si_preference | str_enum | Desired sum insured range: <b>1_2L</b> (micro), <b>3_5L</b> (standard), <b>10_25L</b> (comprehensive), <b>50L_plus</b> (premium). <b>Plan-level input</b> — used to select tier within a confirmed product, not to choose between products.                                  | All 20           | 🟢 Ask 9th |

**Budget band → products:**



| Band    | Annual premium range | Products                                                                           |
|---------|----------------------|------------------------------------------------------------------------------------|
| micro   | < ₹2,000             | SP005, SP019                                                                       |
| budget  | ₹2,000 - ₹10,000     | SP001, SP004, SP009, SP010, SP011, SP012, SP017                                    |
| mid     | ₹10,000 - ₹30,000    | SP001, SP002, SP006, SP007, SP008, SP010, SP013, SP014, SP015, SP016, SP017, SP018 |
| premium | > ₹30,000            | SP003, SP006, SP015, SP016, SP018, SP020                                           |

3.4 Plan-Level Attributes

Asked only after the product is confirmed. These select the tier within the product.

| Attribute                              | Type                   | Definition                                                                                                                                                                                                | Products Covered | Priority              |
|----------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------|
| <code>family_cover</code>              | <code>str_enum</code>  | <code>individual</code> (applicant only), <code>floater_nuclear</code> (2-4 members: self + spouse + children), <code>floater_joint</code> (5-8 members across 3 generations → strongly points to SP020). | All 20           | ● Ask 8th             |
| <code>family_size</code>               | <code>int</code>       | Total number of people to cover including applicant (1-8). Triggers SP020 for 5+ members; affects floater premium loading. Skip if <code>family_cover=individual</code> .                                 | All 20           | ● Ask 8th (follow-up) |
| <code>si_inr</code> (plan)             | <code>int</code> (INR) | The sum insured for a specific plan tier. Ranges from ₹1L (SP005 micro) to ₹2Cr (SP003 global). <code>None</code> for fixed-benefit products (SP011 daily cash, SP020 uses ISI+pool structure).           | 16/20            | ● Plan-level          |
| <code>annual_premium_inr</code> (plan) | <code>int</code> (INR) | Base annual premium for the plan at the sample profile. Excludes GST (18% applicable). Actual premium varies with age, zone, smoker status, occupation (SP012), risk tier (SP015).                        | All 20           | ● Plan-level          |
| <code>opd_limit_inr</code> (plan)      | <code>int</code> (INR) | Annual OPD reimbursement cap for the plan. Ranges ₹3,000 (SP019 Lite) to ₹50,000 (SP018 Platinum). <code>None</code> if no OPD cover.                                                                     | 11/20            | ● Plan-level          |
| <code>deductible_inr</code> (plan)     | <code>int</code> (INR) | Aggregate annual deductible for SP010 (Super Top-Up) plans. Ranges ₹2L-₹25L. The insured absorbs all costs up to this threshold per year; SP010 pays above it. <code>None</code> for all other products.  | SP010 only       | ● Unique to SP010     |

| Attribute                                  | Type                   | Definition                                                                                                      | Products Covered    | Priority          |
|--------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------|-------------------|
| <code>daily_benefit_inr</code> (plan)      | <code>int</code> (INR) | Fixed daily cash amount for SP011 (Hospital Daily Cash) plans. ₹1,000–₹5,000/day. ICU benefit = 2× this amount. | SP011 only          | ● Unique to SP011 |
| <code>maternity_normal_inr</code> (plan)   | <code>int</code> (INR) | Normal delivery reimbursement limit. ₹25,000 (SP017 Basic) to ₹2,00,000 (SP016 Elite).                          | SP007, SP016, SP017 | ●                 |
| <code>maternity_csection_inr</code> (plan) | <code>int</code> (INR) | C-section limit. Always 150% of normal delivery limit.                                                          | SP007, SP016, SP017 | ●                 |
|                                            |                        |                                                                                                                 |                     |                   |
|                                            |                        |                                                                                                                 |                     |                   |

### 3.5 Core Indemnity Attributes

These describe what the hospitalisation cover actually includes. Most are policy-level constants — the same across all plan tiers within a product.

| Attribute                        | Type                       | Definition                                                                                                                                                                                                                                                | Coverage | Priority |
|----------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|
| <code>no_room_rent_cap</code>    | <code>bool</code>          | Whether any hospital room is covered without a daily cap. If <code>False</code> , room rent is capped (typically 1-2% of SI/day); exceeding the cap triggers proportional deduction on the entire bill, not just the room. <b>True in 13/20 products.</b> | 13/20    | ●        |
| <code>icu_covered</code>         | <code>bool</code>          | Whether ICU charges are covered. Universal across all indemnity products; only absent in pure fixed-benefit products. <b>True in 15/20.</b>                                                                                                               | 15/20    | ○        |
| <code>consumables_covered</code> | <code>bool</code>          | Whether single-use hospital items (gloves, syringes, PPE, IV sets) are covered. Hospitals increasingly charge separately for these; can add 10-20% to a bill. Present in SP002, SP003, SP007, SP015, SP016, SP018, SP020. <b>True in 8/20.</b>            | 8/20     | ●        |
| <code>pre_hosp_days</code>       | <code>int</code><br>(days) | Days before admission for which related expenses (consults, tests, medicines) are reimbursed. Range: 30 (SP004, SP017, SP019) to 60 (SP002, SP003, SP015, SP018).                                                                                         | 15/20    | ○        |
| <code>post_hosp_days</code>      | <code>int</code><br>(days) | Days after discharge for which follow-up expenses are covered. Range: 60 (SP004, SP017, SP019) to 180 (SP002, SP003, SP018). SP015 offers 120 days recognising chronic illness recovery.                                                                  | 15/20    | ○        |
| <code>domiciliary_covered</code> | <code>bool</code>          | Whether treatment at home (when patient cannot be moved to hospital) counts as hospitalisation. Important for elderly/immobile patients. <b>True</b>                                                                                                      | 7/20     | ●        |

| Attribute             | Type           | Definition                                                                                                                                                                                      | Coverage | Priority |
|-----------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|
|                       |                | <b>in 7/20 — SP002, SP006, SP010, SP016, SP018, SP019 (partial), SP020.</b>                                                                                                                     |          |          |
| domiciliary_limit_inr | int<br>(INR)   | Annual cap on domiciliary hospitalisation claims. ₹10,000 (SP017) to ₹100,000 (SP003). Relevant for SP006/SP020 senior members.                                                                 | 7/20     | ●        |
| road_ambulance_inr    | int<br>(INR)   | Emergency road ambulance reimbursement per event. Range: ₹1,000 (SP019) to ₹5,000 (SP002/SP003).                                                                                                | 16/20    | ○        |
| air_ambulance_covered | bool           | Whether helicopter/air evacuation costs are covered. Present in SP002 (₹5L), SP003 (₹10L/hosp), SP016 Elite (₹2L), SP018 (₹5L), SP020 (₹3L). <b>True in 5/20.</b>                               | 5/20     | ●        |
| international_cover   | bool           | Whether hospitalisation outside India is covered. <b>Only SP003.</b> Also SP012 (accident only, worldwide). Points immediately to SP003 when <b>True</b> .                                      | 2/20     | ●        |
| ayush_pct_si          | float<br>(0-1) | Proportion of SI available for Ayurveda, Yoga, Unani, Siddha, Homeopathy treatment. Distribution: 0% (SP007), 25% (9 products), 50% (5 products), 100% (SP003), N/A (4 fixed-benefit products). | 16/20    | ○        |
| cashless_hospitals    | int            | Size of empanelled hospital network for cashless treatment. Values: 0 (fixed-benefit: SP011, SP012, SP013, SP014), 5,000 (SP005), 12,000 (8 products), 14,000 (7 products).                     | 18/20    | ○        |
| daycare_procedures    | int            | Number of day-care procedures covered (IRDAI standardised                                                                                                                                       | 18/20    | ○        |

| Attribute | Type | Definition                                                        | Coverage | Priority |
|-----------|------|-------------------------------------------------------------------|----------|----------|
|           |      | list). 541 (full list, most products),<br>100 (SP019 simplified). |          |          |
|           |      |                                                                   |          |          |
|           |      |                                                                   |          |          |

3.6 Waiting Period Attributes

| Attribute            | Type            | Definition                                                                                                                                                                                                                       | Distribution                                   | Priorit                                                                               |
|----------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| waiting.initial_days | int<br>(days)   | Days from inception before any illness claim is accepted. Accidents always exempt. Values: 0 (SP012, SP016, SP019, SP020), 30 (most), 90 (SP013, SP014 — critical illness products).                                             | 0d: 4 products · 30d: 14 · 90d: 2              |    |
| waiting.ped_months   | int<br>(months) | Months before a pre-existing disease is covered. Critical decision driver. Values: 0 (SP016 Day 1), 12 (SP005, SP006, SP015, SP017), 24 (SP009, SP011), 36 (SP002, SP003, SP007, SP008, SP010, SP018), 48 (SP001, SP004, SP013). | 0mo: 2 · 12mo: 4 · 24mo: 2 · 36mo: 6 · 48mo: 3 |  |

| Attribute                                        | Type                         | Definition                                                                                                                                                                                                                           | Distribution    | Priorit |
|--------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|
| <code>waiting.ped_months_diabetes_cardiac</code> | <code>int</code><br>(months) | Reduced PED wait specifically for diabetes, hypertension, and cardiac conditions.<br><b>Only SP015 — 12 months.</b><br>This is the key differentiator that routes diabetic/cardiac users away from standard products (36–48 months). | SP015 only      | ●       |
| <code>waiting.maternity_months</code>            | <code>int</code><br>(months) | Waiting period before maternity benefits activate. 0 (SP016, SP017 — group, Day 1), 9 months (SP007, SP020). Not applicable elsewhere.                                                                                               | 0mo: 2 · 9mo: 2 | ●       |
| <code>waiting.accident_exempt</code>             | <code>bool</code>            | Whether accidents are exempt from initial waiting period. <code>True</code> for all 20 products.                                                                                                                                     | 20/20           | ○       |

PED waiting period distribution:

| Wait             | Products                                                               |
|------------------|------------------------------------------------------------------------|
| Day 1 (0 months) | SP016, SP017 (group — mandated by IRDAI)                               |
| 12 months        | SP005, SP006, SP015 (also diabetes/cardiac on SP015)                   |
| 24 months        | SP009, SP011                                                           |
| 36 months        | SP002, SP003, SP007, SP008, SP010, SP018                               |
| 48 months        | SP001, SP004, SP013                                                    |
| N/A              | SP012 (accident only), SP014 (cancer — prior cancer excluded entirely) |

### 3.7 NCB (No Claim Bonus) Attributes

| Attribute                     | Type                  | Definition                                                                                                                                                                                                                 | Distribution                                           | Priority |
|-------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------|
| <code>ncb.type</code>         | <code>str_enum</code> | How the policy rewards claim-free years. <code>si_increase</code> — sum insured grows (more cover, same price); <code>premium_discount</code> — renewal premium reduces (same cover, cheaper); <code>none</code> — no NCB. | SI increase: 10<br>· Premium<br>discount: 3<br>None: 7 | ●        |
| <code>ncb.pct_per_year</code> | <code>float</code>    | Rate of increase/discount per consecutive claim-free year. 5% (SP004, SP006, SP015), 10% (SP001, SP002, SP003, SP008, SP009, SP018, SP020).                                                                                | Range: 5%-<br>10%                                      | ●        |
| <code>ncb.max_pct</code>      | <code>float</code>    | Maximum accumulated NCB. 20% (SP011, SP012), 25% (SP006), 30% (SP010), 50% (SP001, SP002, SP003, SP008, SP009, SP018, SP020).                                                                                              | Range: 20%-<br>50%                                     | ○        |



### 3.8 Co-pay and Deductible Attributes

| Attribute                         | Type                  | Definition                                                                                                                                                                                                                                                | Coverage            | Priority                                                                            |
|-----------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|
| <code>copay_type</code>           | <code>str_enum</code> | <code>none</code> (insurer pays 100%);<br><code>flat_pct</code> (fixed % on every claim — SP004: 5%);<br><code>age_based_pct</code> (% increases with age — SP006: 10% at 60-69, 20% at 70+).                                                             | Copay applies: 2/20 |  |
| <code>copay_pct</code>            | <code>float</code>    | Flat co-pay percentage. Only SP004 (0.05 = 5%). <code>None</code> for age-based or no co-pay.                                                                                                                                                             | SP004 only          |  |
| <code>aggregate_deductible</code> | <code>bool</code>     | Whether the product operates on an aggregate annual deductible (the insured absorbs all costs up to a threshold per policy year across all hospitalisations).<br><b>Unique to SP010.</b> The key distinction from a per-event deductible (simple top-up). | SP010 only          |  |

### 3.9 Benefit / Fixed-Pay Attributes

These are features beyond standard indemnity — fixed benefit payouts, supplementary income, or condition-specific covers.

| Attribute               | Type      | Definition                                                                                                                                                                                                                | Coverage         |
|-------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| opd_covered             | bool      | Outpatient department cover — consultations, diagnostics, and medicines without hospitalisation. <b>True in 11/20.</b> Absent in SP001, SP004, SP005, SP006, SP010, SP011, SP012, SP013, SP014.                           | 11/20            |
| opd_limit_inr           | int (INR) | Annual OPD reimbursement ceiling. ₹3,000 (SP019 Lite) to ₹50,000 (SP003/SP018 Platinum).                                                                                                                                  | 11/20            |
| maternity_covered       | bool      | Pregnancy-related expenses including delivery, pre/post-natal OPD, newborn cover. <b>True in 4/20: SP007, SP016, SP017, SP020.</b>                                                                                        | 4/20             |
| maternity_day1          | bool      | Whether maternity is covered from Day 1 with no waiting period. <b>Only group products: SP016, SP017.</b> IRDAI-mandated feature of group insurance. Individual policies require 9-month wait.                            | 2/20             |
| newborn_covered         | bool      | Whether the newborn is covered from birth (Day 0) for a defined period (typically 90 days). Includes NICU. Present wherever maternity is covered.                                                                         | 4/20             |
| ivf_covered             | bool      | IVF and assisted reproductive technology. <b>Only SP007 Plan B</b> (₹2L, 3 attempts). SP016 Elite offers it as an optional rider. No product covers it as standard in individual plans beyond Plan B.                     | Effectively 1/20 |
| mental_health_inpatient | bool      | In-patient psychiatric hospitalisation on par with physical illness (Mental Healthcare Act 2017 compliance). Absent only in micro products (SP005, SP019) and pure accident/fixed-benefit products. <b>True in 14/20.</b> | 14/20            |

| Attribute                | Type | Definition                                                                                                                                                                                                                                    | Coverage   |
|--------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| mental_health_opd        | bool | Outpatient therapy sessions (psychologist/psychiatrist) within OPD benefit. SP008 leads at ₹50,000/year (12 sessions). <b>True in 7/20: SP002, SP008, SP009, SP016, SP018, SP020.</b>                                                         | 7/20       |
| critical_illness_lumpsum | bool | Fixed lump-sum on diagnosis of a listed critical illness, regardless of actual treatment cost. Core feature of SP013 (20 conditions, Tier A/B structure). Also present in SP014. <b>True in 2/20.</b>                                         | 2/20       |
| cancer_stage_benefit     | bool | Stage-based payout: CIS=25% SI, Stage I=50%, Stage II-IV=100%, plus monthly income protection for up to 24 months. <b>SP014 only.</b>                                                                                                         | SP014 only |
| income_protection        | bool | Monthly payment during active treatment or disability. SP014: ₹10-30K/month for up to 24 months (cancer treatment). SP012: 1% CSI/week for up to 104 weeks (accident TTD). SP011: daily cash implicitly replaces income. <b>True in 3/20.</b> | 3/20       |
| hospital_daily_cash      | bool | Fixed daily cash per hospitalisation day, no bills needed — only discharge summary. <b>SP011 only.</b> Can be claimed alongside any indemnity policy. ICU benefit = 2× daily rate.                                                            | SP011 only |
| accidental_death_benefit | bool | 100% of Capital Sum Insured paid to nominee on accidental death. Core in SP012. Built-in at ₹1L in SP019. <b>True in 2/20.</b>                                                                                                                | 2/20       |
| disability_benefit       | bool | Permanent total disability (100% CSI), permanent partial disability (per schedule, 5-70% CSI), temporary total disability (1%                                                                                                                 | SP012 only |

| Attribute                  | Type     | Definition                                                                                                                                                                                                                            | Coverage   |
|----------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                            |          | CSI/week, max 104 weeks). <b>SP012 only.</b>                                                                                                                                                                                          |            |
| women_specific_cancer_day1 | bool     | Breast, cervical, ovarian, and uterine cancers covered from Day 1 without waiting period. Annual mammogram + Pap smear + HPV test included. <b>SP008 only.</b>                                                                        | SP008 only |
| restoration_type           | str_enum | How sum insured is replenished after mid-year exhaustion.<br>(once_different_illness) (8 products),<br>(unlimited_different_illness) (SP002, SP003, SP018 — strongest form), (none) (5 products), N/A for fixed-benefit products (4). | All 20     |

### 3.10 Digital / Technology Attributes

| Attribute         | Type | Definition                                                                                                                                                                                                             | Coverage   | Priority               |
|-------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------|
| wearable_discount | bool | Activity data from fitness trackers (Apple Watch, Fitbit, Garmin etc.) linked to premium discount at renewal. SP018: up to 30% (Platinum = 90-100 health score). SP009: up to 10% (step targets). <b>True in 2/20.</b> | 2/20       | ●<br>(points to SP018) |
| ai_health_engine  | bool | SwasthyaAI™ platform — weekly health score (0-100) from wearable data, personalised nudges, AI-assisted claims. <b>SP018 only.</b>                                                                                     | SP018 only | ●                      |
| genetic_testing   | bool | Free genomic panel at enrolment (cardiovascular, cancer, pharmacogenomics, diabetes markers). Genetic data never used for claims or                                                                                    | SP018 only | ●                      |

| Attribute                      | Type        | Definition                                                                                                                                                                                                   | Coverage   | Priority                                                                              |
|--------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------|
|                                |             | premium revision. <b>SP018 only.</b>                                                                                                                                                                         |            |                                                                                       |
| <b>cdmp</b>                    | <b>bool</b> | Chronic Disease Management Programme — dedicated care coordinator, medication reminders, quarterly biometric reviews. SP006 (seniors), SP015 (diabetes/cardiac), SP020 (G1 generation). <b>True in 3/20.</b> | 3/20       |    |
| <b>telemedicine</b>            | <b>bool</b> | Free unlimited telemedicine via app. Present in SP002, SP003, SP006, SP007 (partial), SP008, SP009, SP015-SP020.                                                                                             | 10/20      |    |
| <b>aadhaar_only_enrollment</b> | <b>bool</b> | Policy purchasable with Aadhaar alone — no bank statements, salary slips, or medical forms. <b>SP005 and SP019.</b> Critical for informally employed.                                                        | 2/20       |    |
| <b>pmjay_coordination</b>      | <b>bool</b> | Explicitly designed to supplement Ayushman Bharat (PM-JAY) by covering OPD and primary care gaps PM-JAY excludes. <b>SP019 only.</b>                                                                         | SP019 only |  |
| <b>hr_portal</b>               | <b>bool</b> | Employer-facing admin portal with bulk onboarding, HRMS API integration (Darwinbox, SAP, Keka), claims dashboard. SP016 (full), SP017 (simplified). <b>True in 2/20.</b>                                     | 2/20       |  |
| <b>annual_health_checkup</b>   | <b>bool</b> | Comprehensive preventive health check-up included annually at empanelled centres. SP006, SP013, SP015, SP016, SP018, SP020. <b>True in 6/20.</b>                                                             | 6/20       |  |

| Attribute            | Type  | Definition                                                                                                                                                                                               | Coverage   | Priority |
|----------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| section_80d          | bool  | Premium qualifies for income tax deduction under Section 80D. <b>True in 19/20.</b> Only SP012 (Personal Accident) does NOT qualify — it is not classified as health insurance under the Income Tax Act. | 19/20      | ○        |
| auto_si_increase_pct | float | Annual automatic SI increase per member without fresh underwriting. <b>SP020 only — 5%/year.</b> At 5% ASI, SI doubles in ~14 years. Protects against medical inflation.                                 | SP020 only | ●        |

## 4. Coverage Analytics Summary

### 4.1 Feature Prevalence (% of products where True)

| Feature                 | Products (of 20) | %   |
|-------------------------|------------------|-----|
| section_80d             | 19               | 95% |
| mental_health_inpatient | 14               | 70% |
| no_room_rent_cap        | 13               | 65% |
| opd_covered             | 11               | 55% |
| consumables_covered     | 8                | 40% |
| domiciliary_covered     | 7                | 35% |
| mental_health_opd       | 7                | 35% |
| annual_health_checkup   | 6                | 30% |
| air_ambulance_covered   | 5                | 25% |
| cdmp                    | 3                | 15% |
| income_protection       | 3                | 15% |
| maternity_covered       | 4                | 20% |

| Feature                  | Products (of 20) | %       |
|--------------------------|------------------|---------|
| international_cover      | 2                | 10%     |
| wearable_discount        | 2                | 10%     |
| aadhaar_only_enrollment  | 2                | 10%     |
| hr_portal                | 2                | 10%     |
| critical_illness_lumpsum | 2                | 10%     |
| accidental_death_benefit | 2                | 10%     |
| maternity_day1           | 2                | 10%     |
| 14 features              | 1 each           | 5% each |

#### 4.2 Restoration Type Distribution

| Type                        | Products                                                   | Notes                                        |
|-----------------------------|------------------------------------------------------------|----------------------------------------------|
| once_different_illness      | SP001, SP006, SP007, SP008, SP009, SP015, SP016, SP020 (8) | Standard — refills once if different illness |
| unlimited_different_illness | SP002, SP003, SP018 (3)                                    | Best-in-class — unlimited refills            |
| none                        | SP004, SP005, SP011, SP013, SP017 (5)                      | No restoration                               |
| N/A (fixed-benefit)         | SP010, SP012, SP014, SP019 (4)                             | Concept doesn't apply                        |

#### 4.3 NCB Type Distribution

| Type                                 | Products                                        | Effect                 |
|--------------------------------------|-------------------------------------------------|------------------------|
| SI increase (10%/yr, max 50%)        | SP001, SP002, SP003, SP008, SP009, SP018, SP020 | More cover, same price |
| SI increase (5%/yr, max 25–30%)      | SP004, SP006, SP015                             | Slower growth          |
| Premium discount (5%/yr, max 20–30%) | SP010, SP011, SP012                             | Cheaper renewal        |
| None                                 | SP005, SP007, SP013, SP014, SP016, SP017, SP019 | No NCB                 |

4.4 Cashless Network Distribution

| Network size                 | Products                                               |
|------------------------------|--------------------------------------------------------|
| 0 (N/A — fixed benefit)      | SP011, SP012, SP013, SP014                             |
| 5,000 (government hospitals) | SP005                                                  |
| 12,000                       | SP001, SP004, SP006, SP009, SP010, SP015, SP017, SP019 |
| 14,000                       | SP002, SP003, SP007, SP008, SP016, SP018, SP020        |

4.5 AYUSH Coverage Distribution

| % of SI | Products                                                      |
|---------|---------------------------------------------------------------|
| 0%      | SP007                                                         |
| 25%     | SP001, SP004, SP005, SP008, SP009, SP010, SP015, SP017, SP019 |
| 50%     | SP002, SP006, SP016, SP018, SP020                             |
| 100%    | SP003                                                         |
| N/A     | SP011, SP012, SP013, SP014                                    |

5. Question Priority Queue

The 9 askable user schema fields, in priority order, with the filtering impact of each:



| # | Field                                                | Type        | Impact                                                                      | Skip condition                                                        |
|---|------------------------------------------------------|-------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | <code>buyer_type</code>                              | enum        | Eliminates SP016/SP017 for individual; resolves entirely for employer       | Never skip                                                            |
| 2 | <code>age</code>                                     | int         | Eliminates age-restricted products (SP006, SP009, SP007, SP008)             | Never skip                                                            |
| 3 | <code>gender</code>                                  | enum        | Unlocks SP007/SP008 (female); eliminates them (male)                        | Skip if clearly implied                                               |
| 4 | <code>primary_need</code>                            | enum        | 7 of 8 values resolve to exactly 1 product                                  | Never skip                                                            |
| 5 | <code>has_ped</code> + <code>ped_type</code>         | bool + enum | <code>diabetes_cardiac</code> → SP015 (12mo wait vs 36-48mo); group = Day 1 | Skip <code>ped_type</code> if <code>has_ped=False</code>              |
| 6 | <code>needs_opd</code>                               | bool        | Eliminates 9 products when <code>True</code>                                | Skip if <code>primary_need</code> already resolves                    |
| 7 | <code>budget_band</code>                             | enum        | Guardrail — reduces score of out-of-band products                           | Never skip                                                            |
| 8 | <code>family_cover</code> + <code>family_size</code> | enum + int  | <code>floater_joint</code> with 5+ members → SP020                          | Skip <code>family_size</code> if <code>family_cover=individual</code> |
| 9 | <code>si_preference</code>                           | enum        | Selects plan tier within confirmed product                                  | Ask only after product resolved                                       |

**Minimum for retrieval to run:** `buyer_type` + `age` + `primary_need` (fields 1, 2, 4)

**Minimum for plan selection:** `resolved_product_id` + one of (`budget_band` or `si_preference`)

## 6. File Inventory

| File                                                              | Purpose                       | Lines             |
|-------------------------------------------------------------------|-------------------------------|-------------------|
| <code>swasthya_SP001.txt</code> - <code>swasthya_SP020.txt</code> | Policy documents (RAG corpus) | ~1,000-1,500 each |

| File                                    | Purpose                                                    | Lines |
|-----------------------------------------|------------------------------------------------------------|-------|
| <code>policy_metadata.json</code>       | Machine-readable product index                             | —     |
| <code>policy_feature_registry.py</code> | Typed Python registry — ground truth for all agent logic   | 2,368 |
| <code>attribute_glossary.py</code>      | 75-attribute definitions with LLM context generation       | 1,808 |
| <code>retrieval_tools.py</code>         | 4 LLM-callable tools: filter, features, plans, regulations | 869   |
| <code>user_schema.py</code>             | Typed UserSchema dataclass with sufficiency gates          | 886   |
| <code>fsm.py</code>                     | FSM states, transitions, sub-agents, question templates    | 1,129 |

**Total codebase: ~9,060 lines of typed, tested Python**

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